陈琳 Lin Chen

Postdoctoral Researcher, Northeastern University

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Profile

Lin Chen is a postdoctoral researcher at the Network Science Institute, Northeastern University, US. In 2025, she obtained her Ph.D. in Computer Science from the Hong Kong University of Science and Technology. In 2020, she obtained her B.S. in Communication Engineering from Tongji University. Her research interests lie in AI agents, urban science, and behavioral science. She has published first-author papers in Nature Human Behaviour, KDD, www, IJCAI, and ICWSM, which have been covered by Science News, Mirage News, Newswise, etc. She has been recognized as one of the Rising Stars Women in Engineering, Asian Deans' Forum.

Research Experience

Postdoctoral Researcher *NetSI, Northeastern University* **Boston, US** *09/2020 - 07/2025*

- Advisor: Prof. Esteban Moro

Visiting Researcher *Future Intelligence Lab, Tsinghua University* **Beijing, CN** *04/2023 - 12/2023*

- Advisor: Prof. Yong Li

Visiting Researcher *Knowledge Lab, University of Chicago* **Chicago, US** *12/2022 - 04/2023*

- Advisor: Prof. James Evans

Education

Ph.D. *Hong Kong University of Science and Technology* **Hong Kong, CN** *09/2020 - 07/2025*

- Major: Computer Science and Engineering
- Advisor: Prof. Pan Hui

BSc *Tongji University* **Shanghai, CN** *09/2016 - 07/2020*

- Major: Communication Engineering

Selected Awards

NetSci Best Poster Award *2026*
Federation of Hong Kong Chiu Chow Community Organizations Fellowship *2025*
Hong Kong PhD Fellowship (HKPFS) *2020-2024*
KDD Student Travel Award *2024*
HKUST Conference Travel Grant *2024*
Rising Stars Women in Engineering, Asian Deans′ Forum *2023*
ICWSM Student Travel Grant *2023*
HKUST Overseas Research Award *2022*
HKUST RedBird PhD Award *2020*
Shanghai Outstanding Bachelor Graduate *2020*
Outstanding Student of Tongji University *2019*

National Undergraduate Scholarship *2017, 2018, 2019*

Publications

Representative Work

5. **Lin Chen**, F. Xu, Z. Han, K. Tang, P. Hui, J. Evans, and Y. Li. Strategic COVID-19 vaccine distribution can simultaneously elevate social utility and equity. *Nature Human Behaviour*, 2022.

4. **Lin Chen**, F. Xu, N. Li, Z. Han, M. Wang, Y. Li, and P. Hui. Large Language Model-driven Meta-structure Discovery in Heterogeneous Information Network. *ACM SIGKDD Conference on Knowledge Discovery and Data Mining (SIGKDD)*, (2024): 307-318. (Acceptance rate: ∼20%)

3. B. Fan∗, **Lin Chen**∗, S. Li, J. Yuan, F. Xu, P. Hui, and Y. Li. Invisible walls in cities: Leveraging large language models to predict urban segregation experience with social media content. *Proceedings of the ACM Web Conference (WWW)* 2026, pp. 8916-8926. (Acceptance rate: 21.7%)

2. **Lin Chen**, Y. Zhang, J. Feng, H. Chai, H. Zhang, B. Fan, Y. Ma, S. Zhang, N. Li, T. Liu, N. Sukiennik, K. Zhao, Y. Li, Z. Liu, F. Xu, and Y. Li. AI Agent Behavioral Science. *Humanities and Social Sciences Communications*, 2026.

1. **Lin Chen**, F. Xu, Q. Hao, P. Hui, and Y. Li. Getting Back on Track: Understanding COVID-19 Impact on Urban Mobility and Segregation with Location Service Data. *International AAAI Conference on Web and Social Media (ICWSM)*, 17 (2023): 126-136. (Acceptance rate: ∼20%)

Under Review/Working Papers

9. **Lin Chen***, Y. Chen*, and Y.Li. Do LLMs Change Their Minds Like Humans? Diagnosing Human−LLM Divergence in Belief Update. Under review in *EMNLP'26*.

8. Y. Liu*, **Lin Chen***, Z. Liu, J. Lu, and D. Murray. LLMs Amplify Citation Bias and Social Proximity by Rhetorical Intent. Under review in *EMNLP'26*.

7. **Lin Chen**, Z. Liu, X. Hu, and Y. Li. AI agents reshape consensus in human groups. Under review in *Science Advances*.

6. **Lin Chen**, F. Xu, E. Moro, P. Hui, Y. Li, and J. Evans. Urban mobility network centrality predicts social resilience. Under revision in *Nature Communications*.

5. **Lin Chen**, G. Weng, and E. Moro. Large language models create an uneven informational layer over cities. Accepted by *IC2S2'26*.

4. Y. Cao, Q. Shi, L. Wang, L. Y. Lo, **Lin Chen**, Y. Han, Y. Wang, K. Chen. SocialFiVis: A Visual Analytics Sandbox for LLM-Driven Multi-Agent Simulation in Social Finance. Under review in *IEEE VIS'26*.

3. F. Xu, C. Shao, **Lin Chen**, Q. Zeng, Z. Chen, N. Sukiennik, J. Wang, C. Gao, H. Wang, J. Lian, X. Xie, Y. Li, and J. Evans. Large Language Models for Computational Social Science: A Survey. Accepted by *Nature Human Behaviour*.

2. F. Xu, Y. Zhang, Z. Zhou, C. Shao, J. Piao, K. Zhao, Z. Chen, Q. Zeng, B. Fan, J. Zhang, Y. Li, X. Liu, R. Zhao, Q. Yang, C. Rong, S. Ren, B. Luo, P. Liu, **Lin Chen**, Y. Yuan, J. Ding, and Y. Li. A Comprehensive Survey of AI Scientists. Under review in *Artificial Intelligence Review*.

1. Q. Hao, **Lin Chen**, X. Qi, Y. Yuan, Z. Zong, H. Chen, K. Zhao, S. Wang, Y. Zhang, J. Yuan, and Y. Li. Reinforcement Learning in the Era of Large Language Models: Challenges and Opportunities. Under review in *ACM Computing Surveys*.

Journal Publications

5. Y. Lin, Z. Zhou, Y. Zheng, H. Wang, J. Ding, F. Xu, C. Gao, Y. Wang, J. Yuan, Q. Gao, **L. Chen**, A. Yu, P. Wang, J. Li, L. Tian, Y. Yue, H. Pan, F. Lu, Z. Lu, Y. Li. From metaphor to model: A complex-systems framework for urban organism. To appear in *Journal of Geo-information Science*, 2026 (Chinese).

4. F. Xu, Q. Wang, E. Moro, **Lin Chen**, A. S. Miranda, M. C. Gonzalez, M. Tizzoni, C. Song, C. Ratti, L. Bettencourt, Y. Li and J. Evans. Mobility Behaviour Data Quantifies Experienced Inequalities in Urban Space. *Nature Human Behaviour*, 2025.

3. Y. Zhang, F. Xu, **Lin Chen**, Y. Yuan, J. Evans, L. Bettencourt, and Y. Li. Counterfactual mobility network embedding reveals prevalent accessibility gaps in US cities. *Humanities and Social Sciences Communications*, 2024.

2. Z. Han, **Lin Chen**, Q. Hao, Q. He, K. Budeski, D. Jin, F. Xu, K. Tang, and Y. Li. How enlightened self-interest guided global vaccine sharing benefits all: A modeling study. *Journal of Global Health*, 2023.

1. Q. Hao, F. Xu, **Lin Chen**, P. Hui and Y. Li. Hierarchical Multi-agent Model for Reinforced Medical Resource Allocation with Imperfect Information. *ACM Transactions on Intelligent Systems and Technology (TIST)*, 2022.

Conference Publications

5. Y. Li, L. Li, **Lin Chen**, Q. Liao, F. Xu, and Y. Li. AgentExpt: Automating AI Experiment Design with LLM-based Resource Retrieval Agent. Accepted by *ICML'26*. (Acceptance rate: 26.6%)

4. **Lin Chen**, Y. Li, and P. Hui. VulnerabilityMap: An Open Framework for Mapping Vulnerability among Urban Disadvantaged Populations in the United States. *Proceedings of the Thirty-Third International Joint Conference on Artificial Intelligence (IJCAI)* AI for Good. Pages 7206-7214. (2024) (Acceptance rate: 25.8%)

3. Q. Hao, F. Xu, **Lin Chen**, P. Hui, and Y. Li. Hierarchical Reinforcement Learning for Scarce Medical Resource Allocation with Imperfect Information. *ACM SIGKDD Conference on Knowledge Discovery and Data Mining (SIGKDD)*, 2021. (Acceptance rate: 21.9%)

2. Q. Hao, **Lin Chen**, F. Xu, and Y. Li. Understanding the Urban Pandemic Spreading of COVID-19 with Real World Mobility Data. *ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (SIGKDD)*, 2020.

1. K. Zhao, J. Feng, Z. Xu, T. Xia, **Lin Chen**, F. Sun, D. Guo, D. Jin, and Y. Li. DeepMM: Deep learning based map matching with data augmentation. *Proceedings of the 27th ACM SIGSPATIAL international conference on advances in geographic information systems (SIGSPATIAL)*. 2019.

Talks and Posters

17. [07/2026, Burlington] "When AI Becomes Urban Compass: Socio-Spatial Biases in LLM-based Place Recommendations." Parallel talk @ the 12th International Conference on Computational Social Science (IC2S2'26). (Soon!)

16. [06/2026, Beijing] "复杂城市系统群体行为建模与干预：从数据驱动到大模型赋能." Invited talk @ Peking University's Young Scholars Forum on Remote Sensing and GIScience.

15. [06/2026, Boston] "When AI Becomes Urban Compass: Socio-Spatial Biases in LLM-based Place Recommendations." Poster @ the 20th International School & Conference on Network Science (NetSci'26)(Best Poster Award).

14. [10/2025, Boston] "Inequality and Resilience in Urban Social Dynamics". Invited talk @ Ryan Wang Lab, Northeastern University.

13. [10/2025, Boston] “Inequality and Resilience in Urban Social Dynamics”. Invited talk @ Network Science Institute, Northeastern University.
12. [09/2025, Siena] “Inequality and Resilience in Urban Social Dynamics”. Invited keynote @ UrbanSys, Conference on Complex Systems (CCS’25).
11. [08/2025, Bali] “Urban mobility network centrality predicts social resilience”. Poster @ the Third Conference on Urban Science and Intelligence.
10. [08/2024, Barcelona] “Large language model-driven meta-structure discovery in heterogeneous information network”, Oral & Poster @ the 30th ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD’24).
9. [08/2024, Beijing] “Mobility network centrality explains urban social resilience during crises”, Poster & Pitch @ the Second Research Summit for Urban Science (RSUS’24).
8. [08/2024, Jeju] “VulnerabilityMap: An Open Framework for Mapping Vulnerability among Urban Disadvantaged Populations in the United States”, Oral & Poster @ the Thirty-Third International Joint Conference on Artificial Intelligence (IJCAI’24).
7. [11/2023, Tokyo] “UrbanLens: Computational Understanding of Urban Lifestyles”, Poster & Pitch @ Rising Stars Women in Engineering Workshop, Asian Deans’ Forum (ADF-RSE’23).
6. [09/2023, Beijing] “Strategic COVID-19 vaccine distribution can simultaneously elevate social utility and equity”, Poster @ the 3rd International Forum on Big Data for Sustainable Development Goals (FBAS’23).
5. [08/2023, Beijing] “Strategic COVID-19 vaccine distribution can simultaneously elevate social utility and equity”, Poster & Pitch @ the First Research Summit for Urban Science (RSUS’23).
4. [06/2023, Limassol] “Getting Back on Track: Understanding COVID-19 Impact on Urban Mobility and Segregation with Location Service Data”, Oral @ the International AAAI Conference on Web and Social Media (ICWSM’23).
3. [01/2023, Chicago] “Complex Network Analysis Reveals Universal Mobility Response to Health Crisis”, Talk @ Knowledge Lab Seminar.
2. [04/2022, Beijing (remote)] “Review on leveraging human mobility data for urban epidemic modeling”, Talk @ Future Intelligence Lab, Tsinghua University.
1. [01/2021, Hong Kong] “COVID-19 Vaccine Distribution: Efficiency and Equity”, Talk @ Fudan-Helsinki-HKUST Webinar.

Participating Projects

- Research on key technologies for whole-life cycle governance of urban organisms, core member, 2024-2027.

Teaching Experience

- COMP 4641: Social Information Network Analysis and Engineering, HKUST, Spring 2021. Teaching Assistant; Instructor: Prof. Pan Hui.
- COMP 4021: Internet Computing, HKUST, Fall 2021. Teaching Assistant; Instructor: Prof. Dik-Lun Lee.

Academic Service

Conference Organizer

- UrbanNetAI Satellite @ NetSci’26, primary organizer, 2026/06.

- The Third Conference on Urban Science and Intelligence, session chair, 2025/08.
- The First Research Summit for Urban Science, session chair, 2023/08.

Conference Reviewer

- ASONAM, TNSE, TheWebConf, IC2S2, IMWUT, WSDM, ICWSM, IJCAI

Journal Reviewer

- Nature, EPJ Data Science, Humanities and Social Sciences Communications (HSSCOMMS), Journal of Social Computing (JSC), Social Network Analysis and Mining